

DATABASES

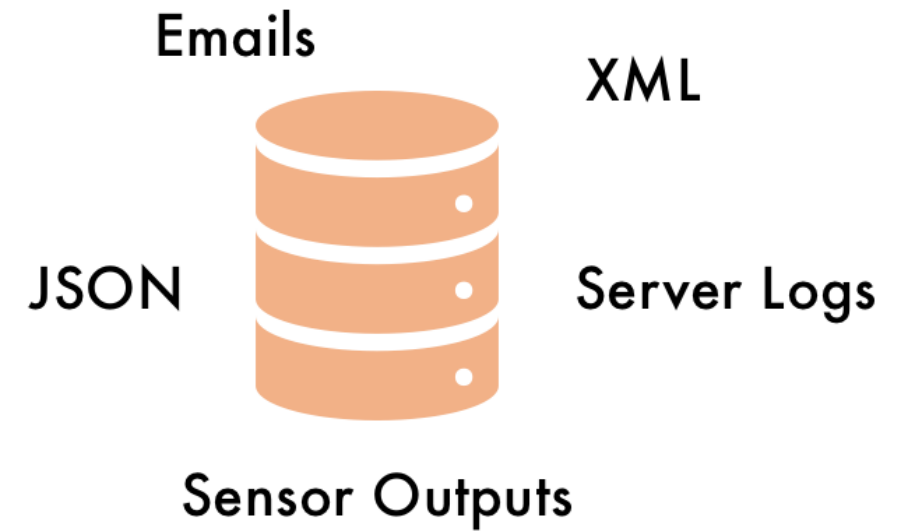
Basic Concepts

TYPES OF DATA

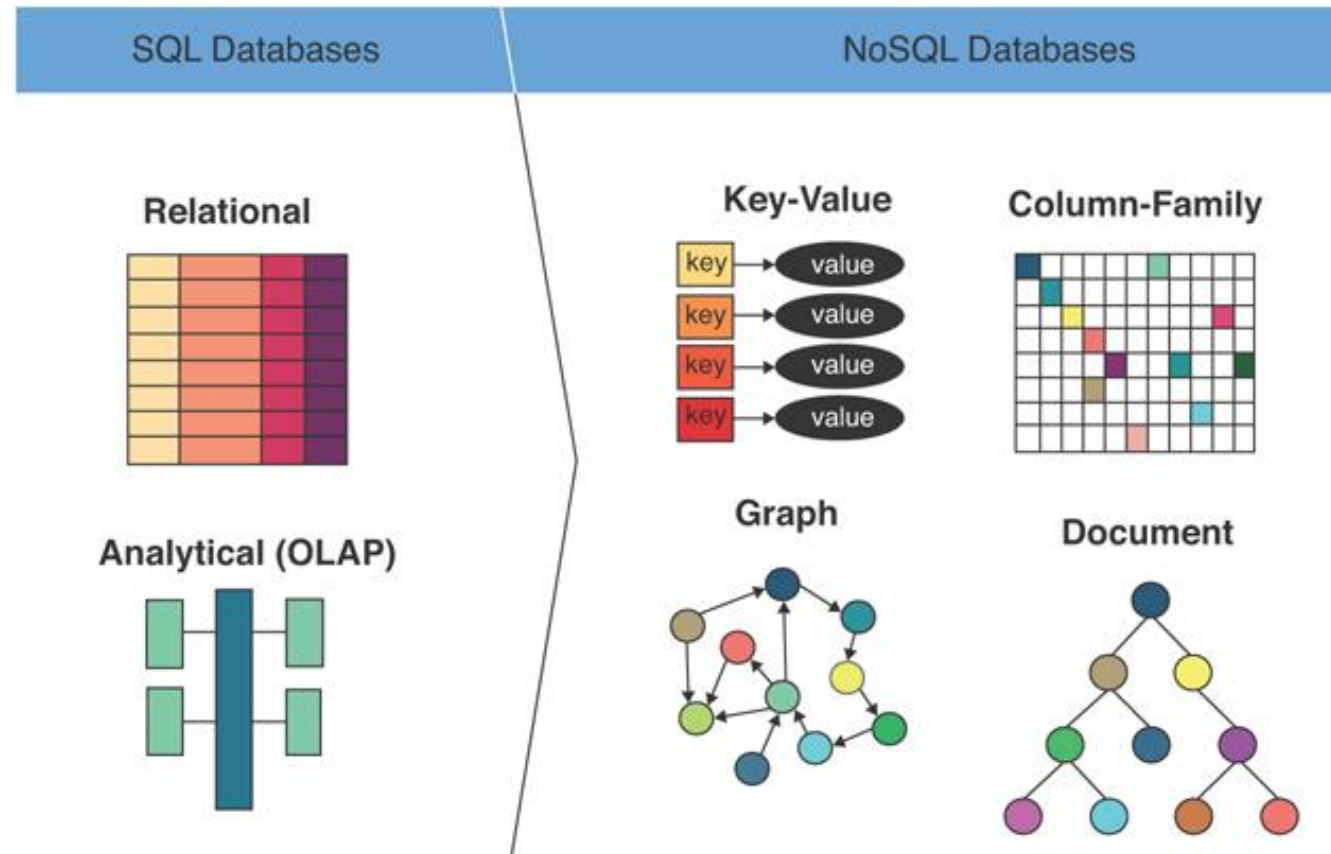
STRUCTURED DATA FORMATS



UNSTRUCTURED DATA FORMATS

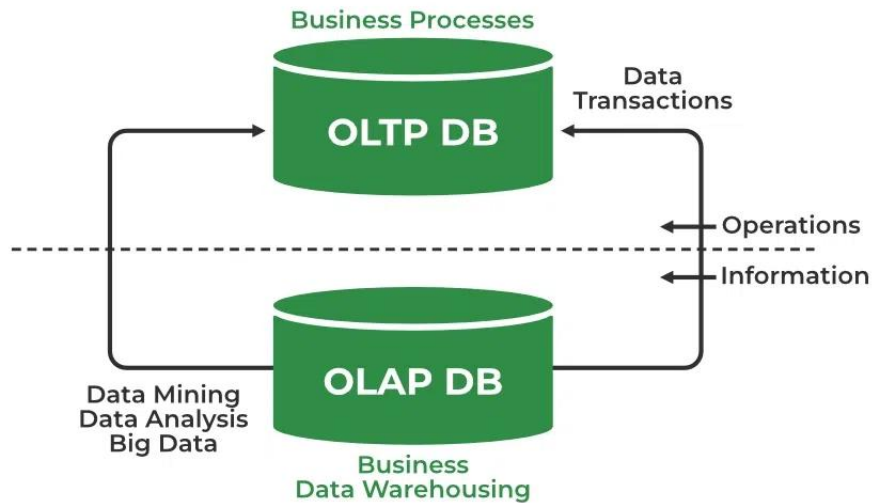


DATA REPOSITORIES



Data repositories: Store, manage, and organize data. Offer a framework for retrieval and administration

OLTP (Online Transaction Processing) VS OLAP (Online Analytical Processing)

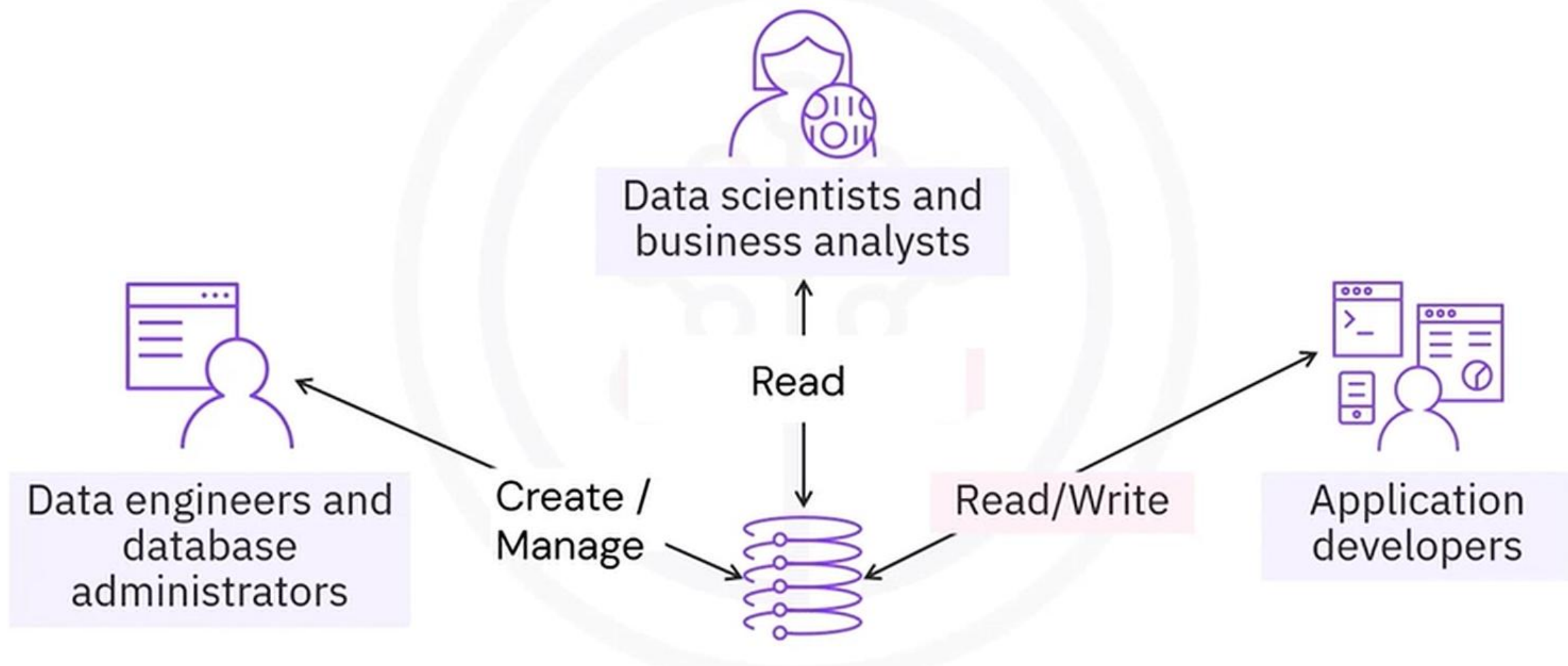


OLTP vs. OLAP

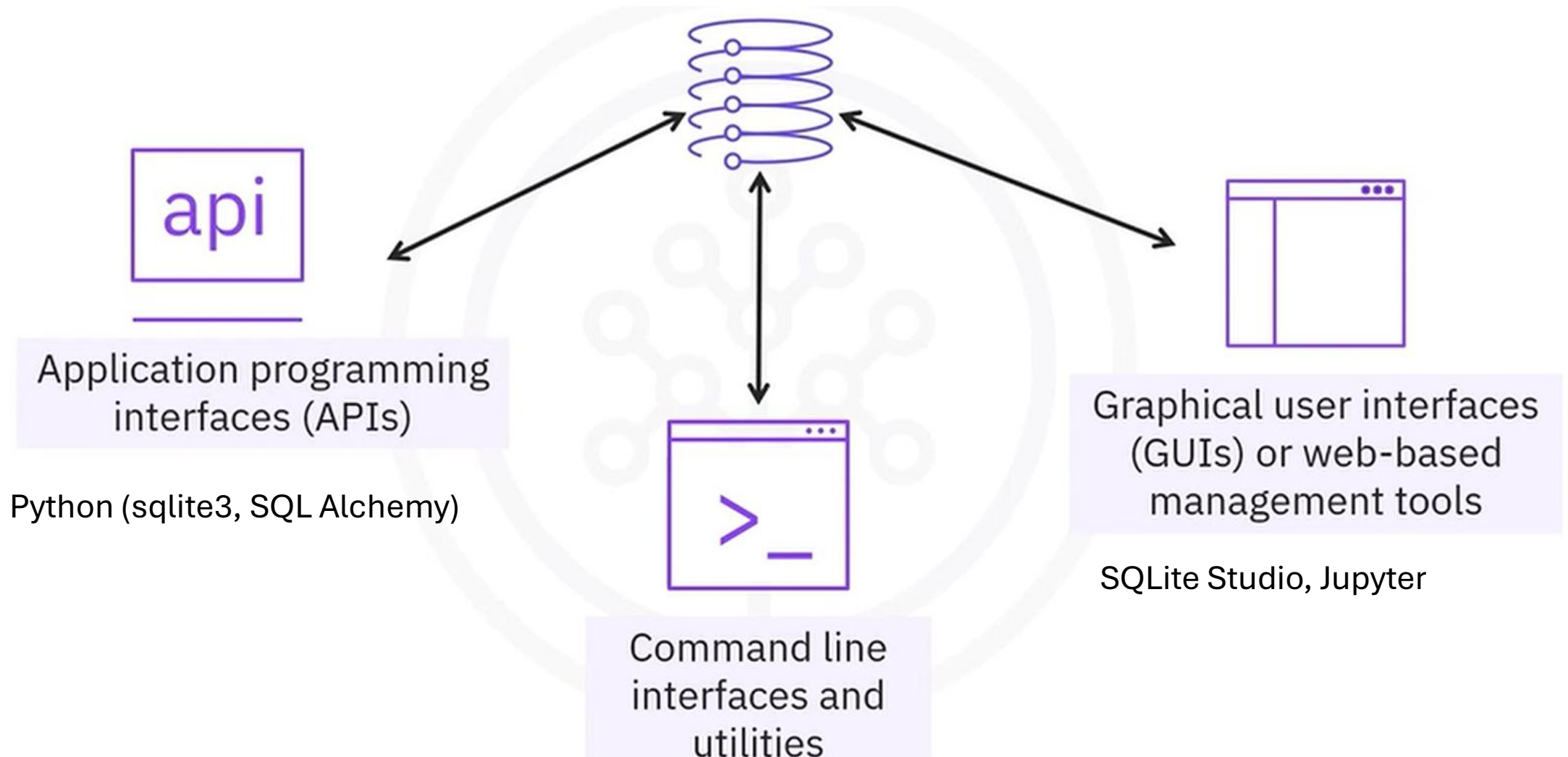
OLTP (Transactional)		OLAP (Analytical)
Goal	Run the business	Analyze the business
Operations	Insert, Update, Delete	Read, Aggregate, Summarize
Data Volume	MB to GB	GB to TB and beyond
Response	Milliseconds	Seconds to minutes
Users	Front-end users	Business analysts, data scientists
Schema	Highly normalized (3NF)	Denormalized (star/snowflake schema)

Key user roles

Primary classes of database users include:



DATABASE TOOLS



Data scientists and business analysts

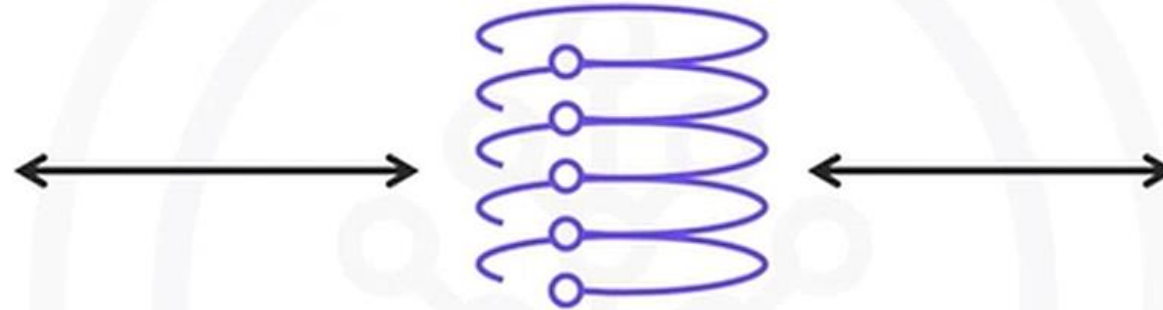


Perform data analysis, insight derivation, and data-driven predictions

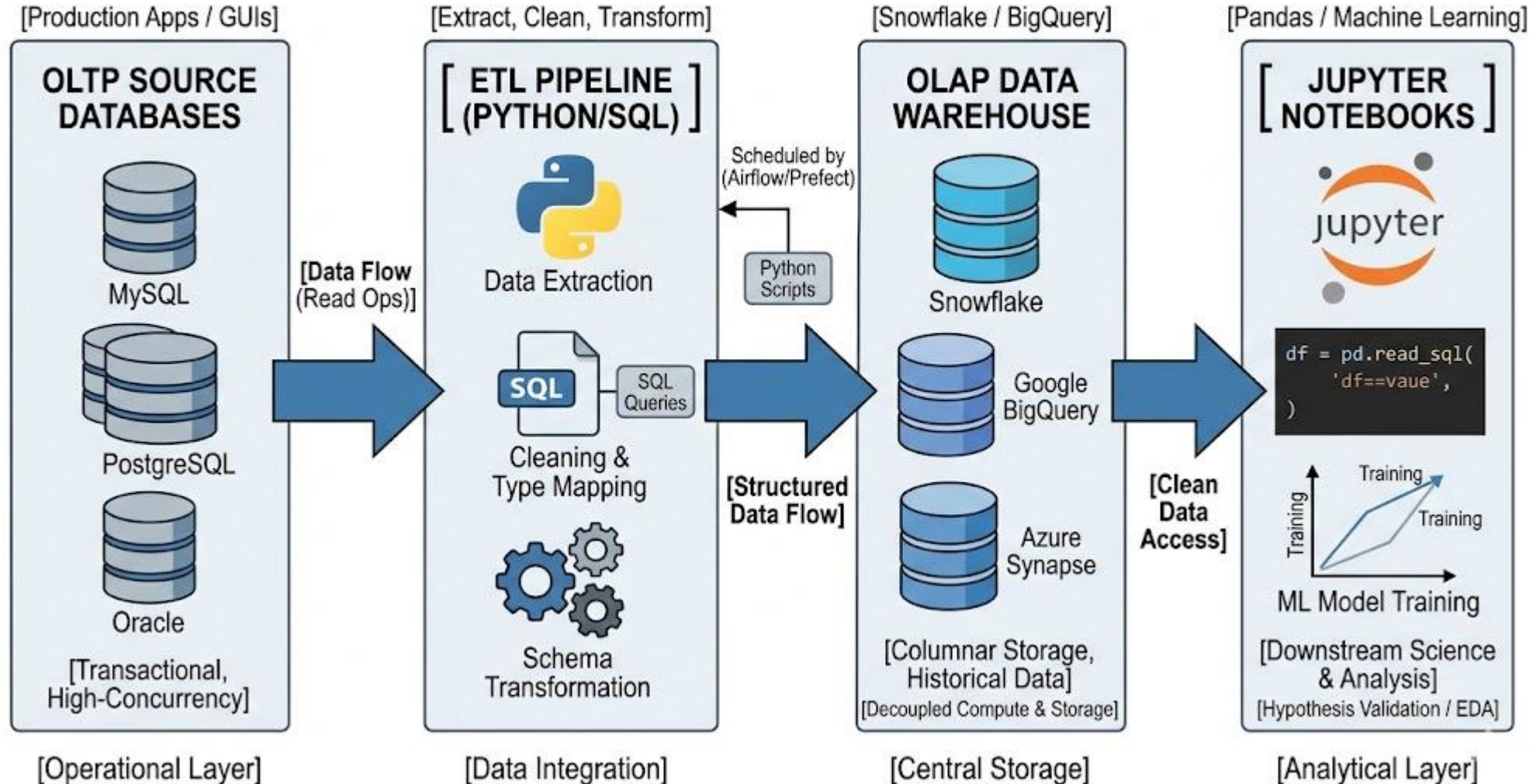
Involve reading from existing data sources

Create and populate database objects occasionally

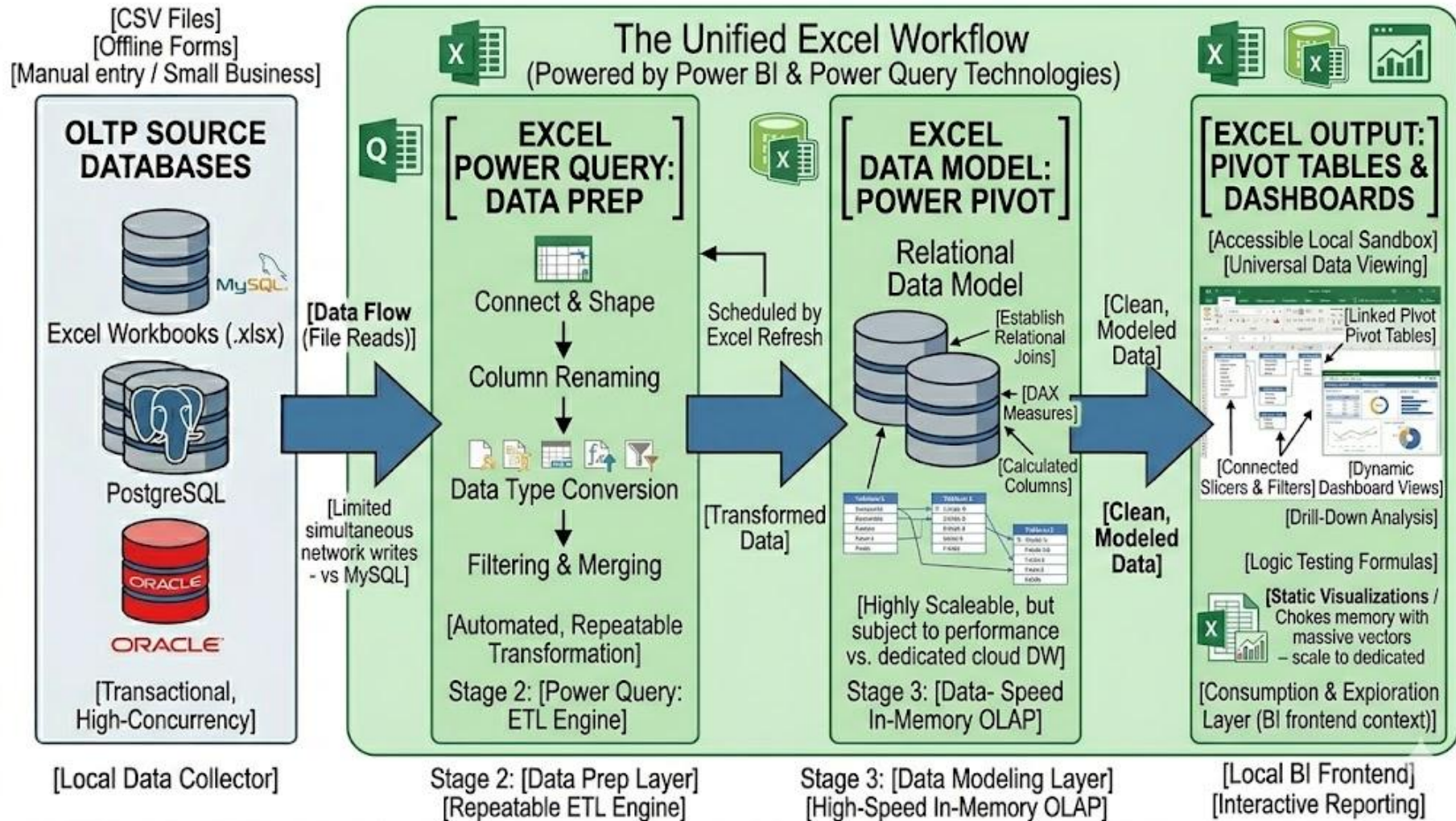
DATA SCIENTIST TOOLS



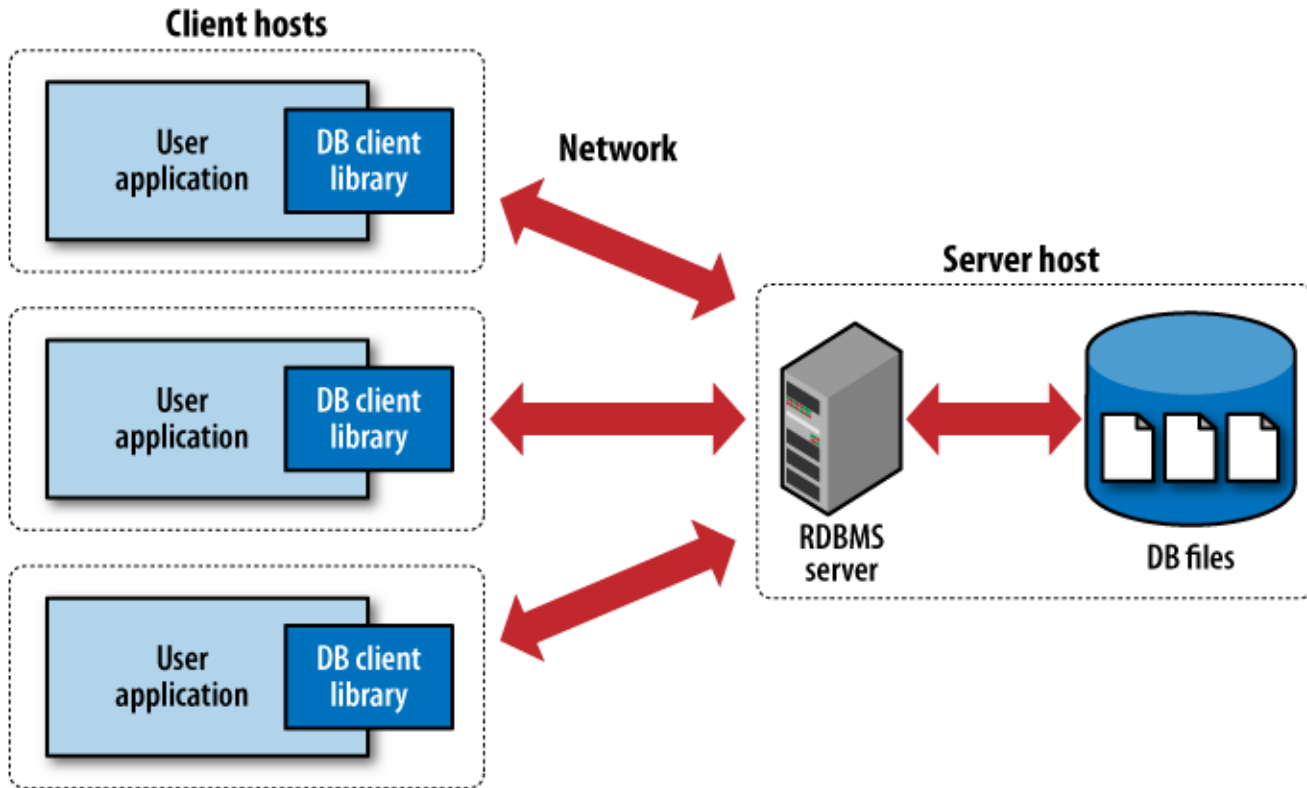
THE DATA SCIENCE PIPELINE



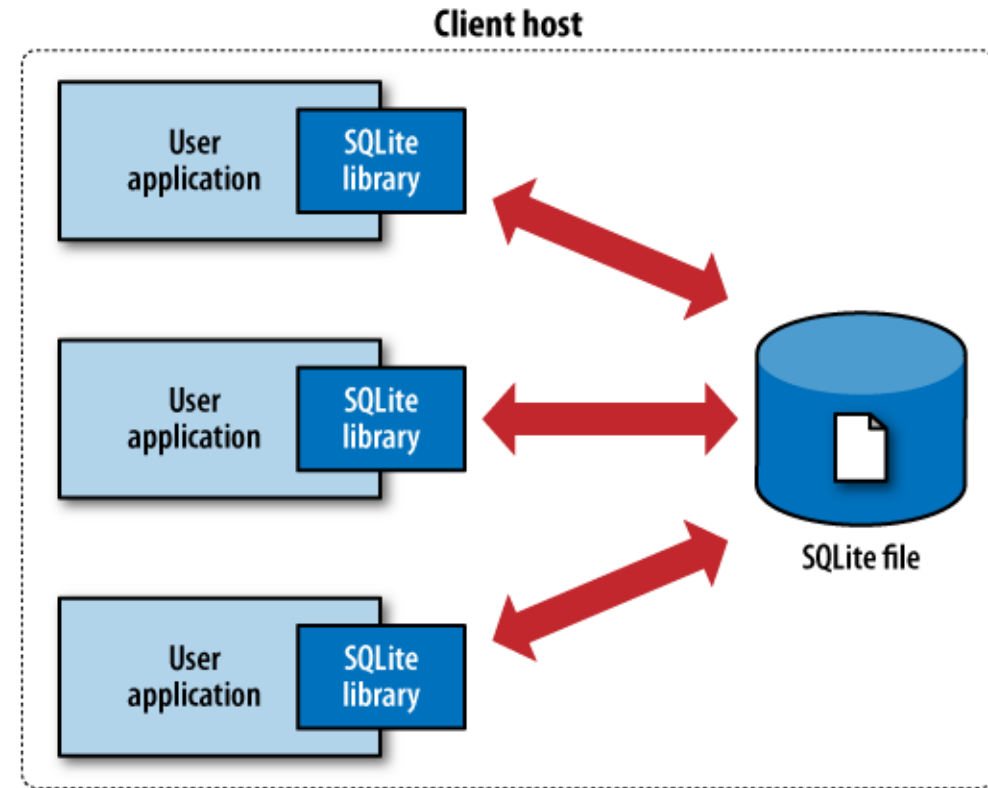
THE DATA SCIENCE PIPELINE



MySQL vs SQLite



(a) Traditional client-server architecture



(b) SQLite serverless architecture

THE DATA SCIENCE PIPELINE

[OLTP: MySQL] → [ETL Pipeline] → [OLAP: Snowflake] → [TABLEAU]
(Captures Data) (Moves Data) (Stores History) (Visualizes Insights)

[Local Data] → [OLTP Operations] → [OLAP Analysis] → [Jupyter Lab]
(.csv / Logs) (Local Write Loops) (Pandas Extract) (Python Modeling)

THE SQLITE SANDBOX

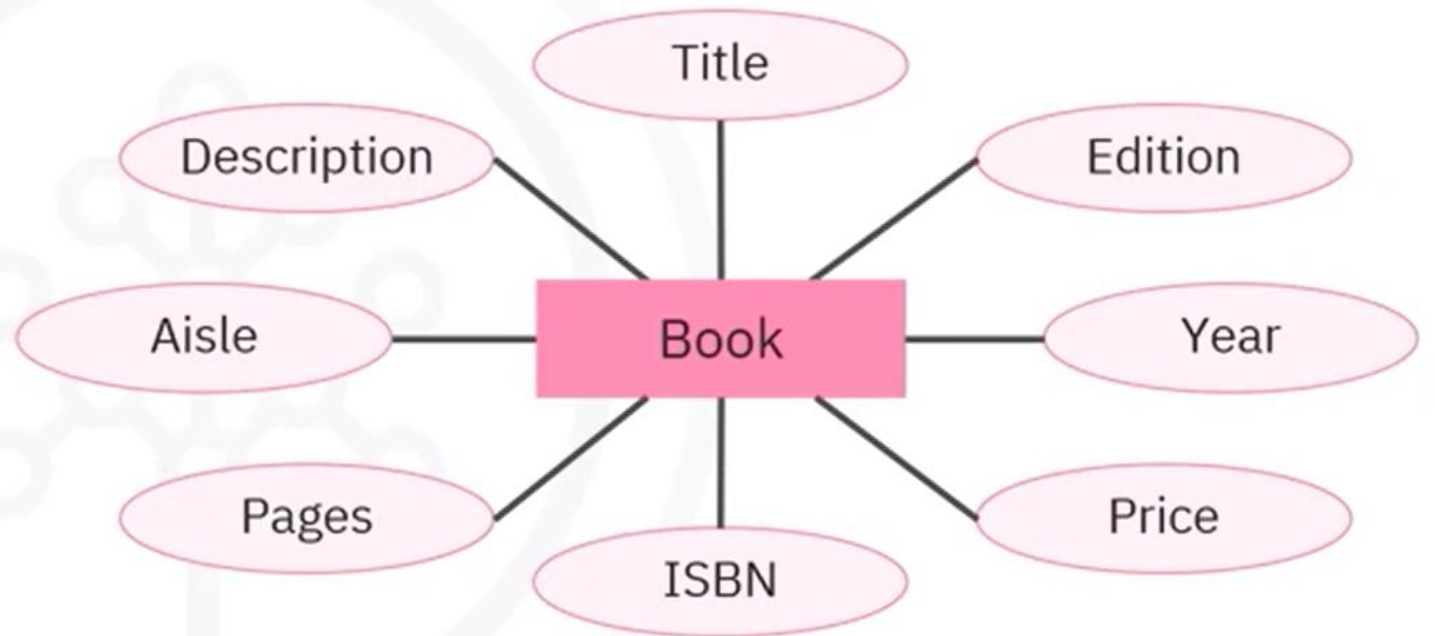
COMPONENTS OF ENTITY-RELATIONSHIP DIAGRAM (ERD)

Entities:

- Represent real-world objects, concepts, or things
- Depicted with rectangles

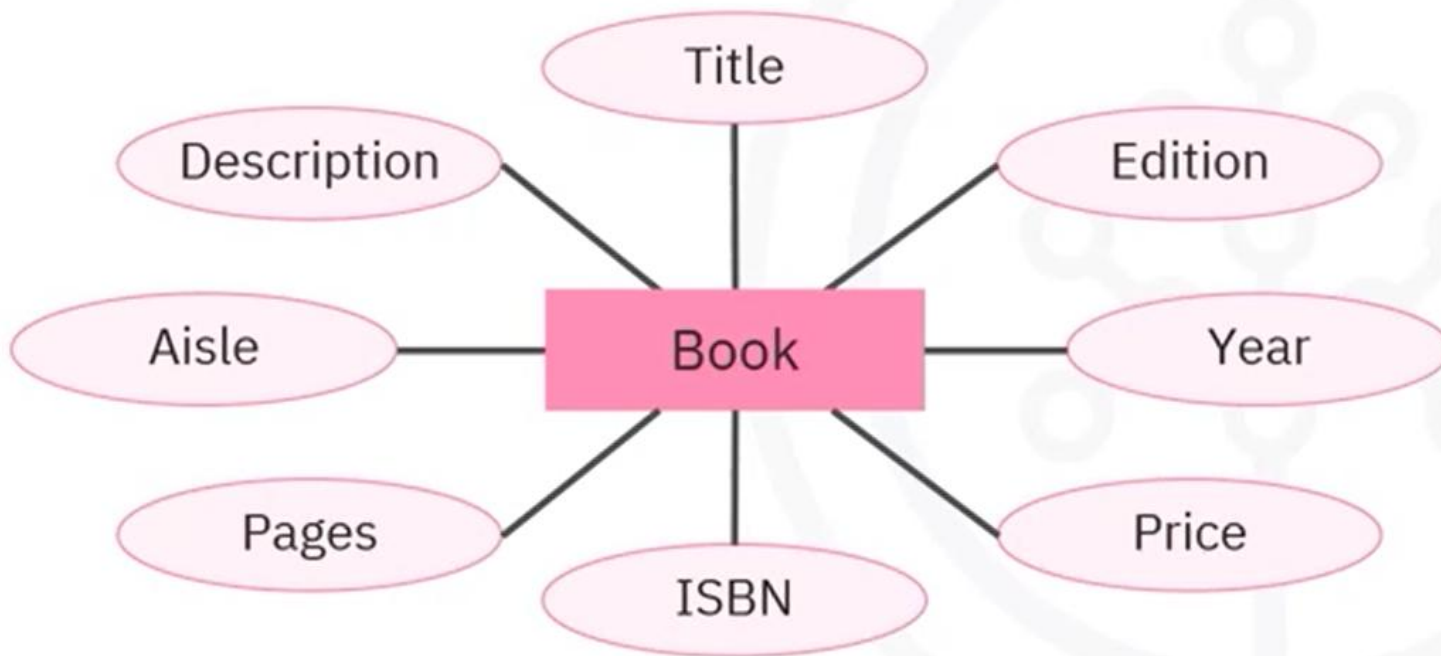
Attributes:

- Denote characteristics or properties of an entity
- Depicted with oval



MAPPING AN ERD INTO A TABLE

Begin with an ERD



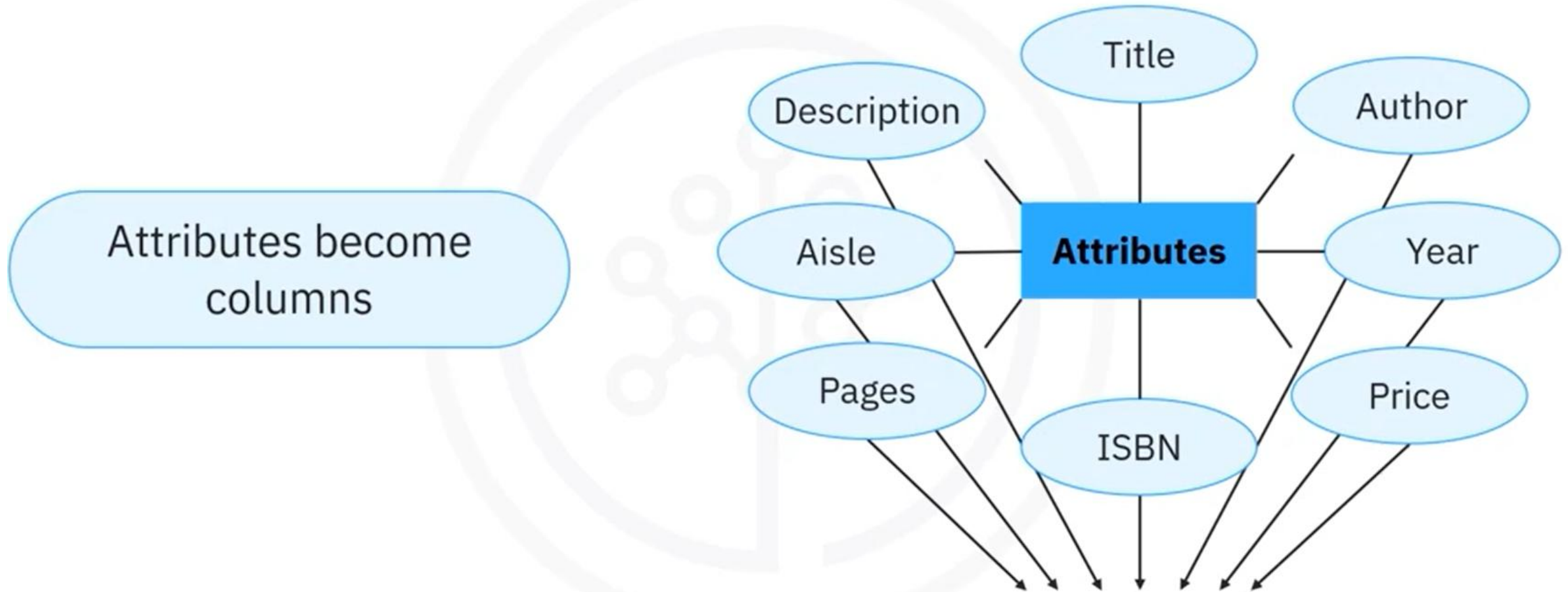
Map the ERD to the tables

Entity



Book

TRANSLATING ATTRIBUTES



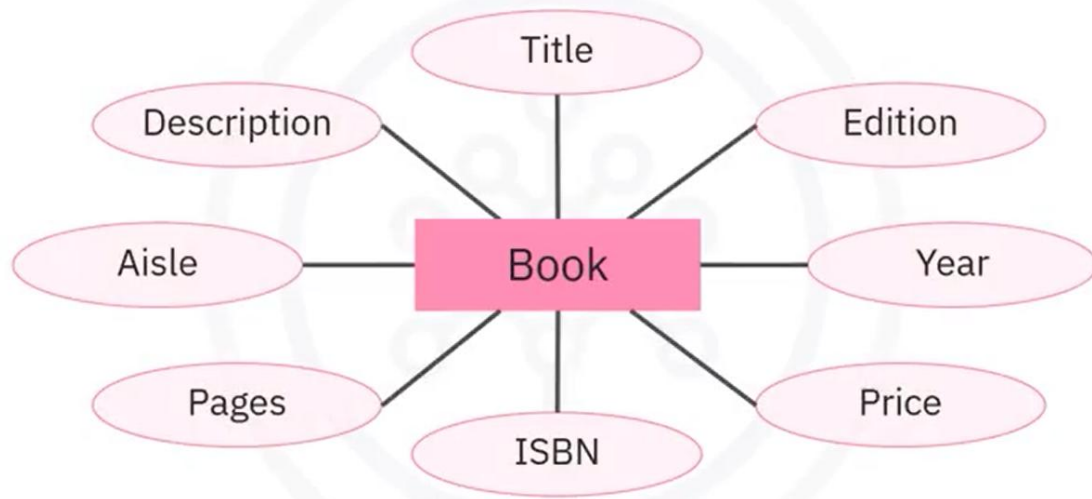
MAPPING AN ERD INTO A TABLE

Books

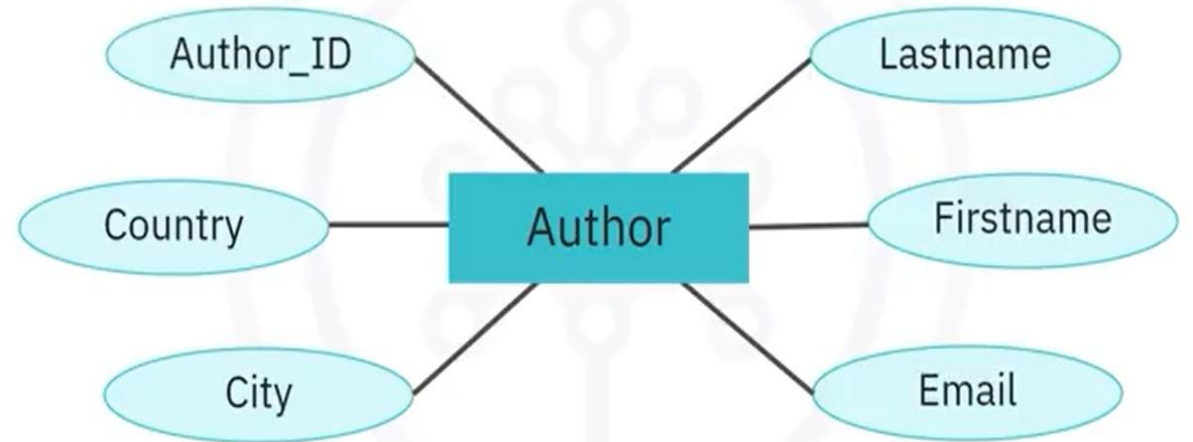
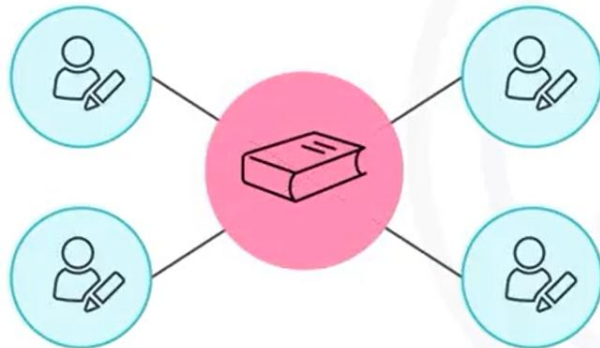
Authors

Last name	First name	Email

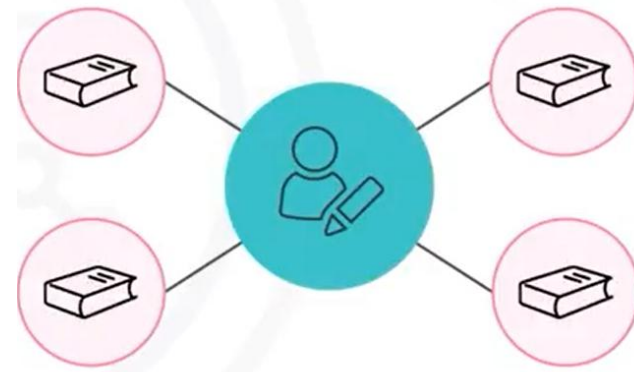
RELATIONSHIPS BETWEEN ENTITIES



Multiple authors
writing a book



One author writing
multiple books



TYPES OF RELATIONSHIPS

One-to-one relationship



- Book written by a single author
- Each entity is engaged in one relationship
- Attributes might clutter the diagram

TYPES OF RELATIONSHIPS

One-to-many relationship



- Book written by multiple authors



- Many-to-one relationship
 - For many authors, writing a single book

TYPES OF RELATIONSHIPS

Many-to-many relationship



- Many authors writing many books

TYPES OF SQL STATEMENTS

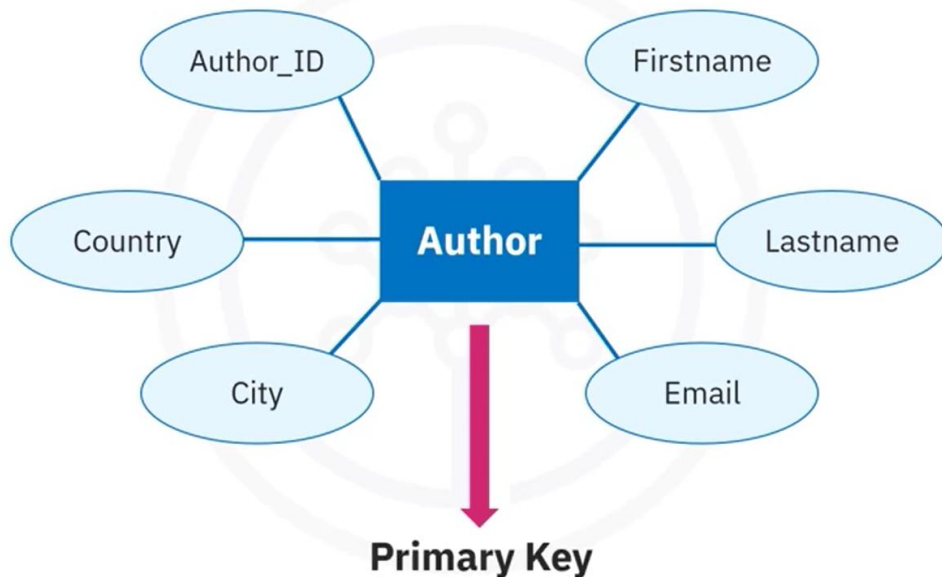
- DDL (Data Definition Language) statements:
 - Define, change, or drop data
- Common DDL:
 - **CREATE:** Used for creating tables and defining their columns.
 - **ALTER:** Adding and dropping columns and modifying their data types.
 - **TRUNCATE:** Used for deleting data in a table, but not the table itself.
 - **DROP:** For deleting tables.

TYPES OF SQL STATEMENTS

- DML (Data Manipulation Language) statements:
 - Read and modify data
 - CRUD operations (**C**reate, **R**ead, **U**ppdate & **D**etele rows)
- Common DML:
 - **INSERT:** Insert a row or rows into a table.
 - **SELECT:** Reads or selects a row or rows from a table.
 - **UPDATE:** Edit a row or rows in a table.
 - **DELETE:** Remove a row or rows from a table.

CREATE TABLE

- To create the Author table, use the following columns and datatypes:
 - AUTHOR(Author_ID:char, Lastname:vvarchar, Firstname:vvarchar, Email:vvarchar, City:vvarchar, Country:char)



```
CREATE TABLE author (  
  Author_ID CHAR(2) PRIMARY KEY NOT NULL,  
  Lastname VARCHAR(15) NOT NULL,  
  Firstname VARCHAR(15) NOT NULL,  
  Email VARCHAR(40),  
  City VARCHAR(15),  
  Country CHAR(2)  
)
```

PRIMARY KEY

- Uniquely identify every row in a table
- Add a column to the table
- Create a primary key across the two columns
- Use the combination of site and employee id

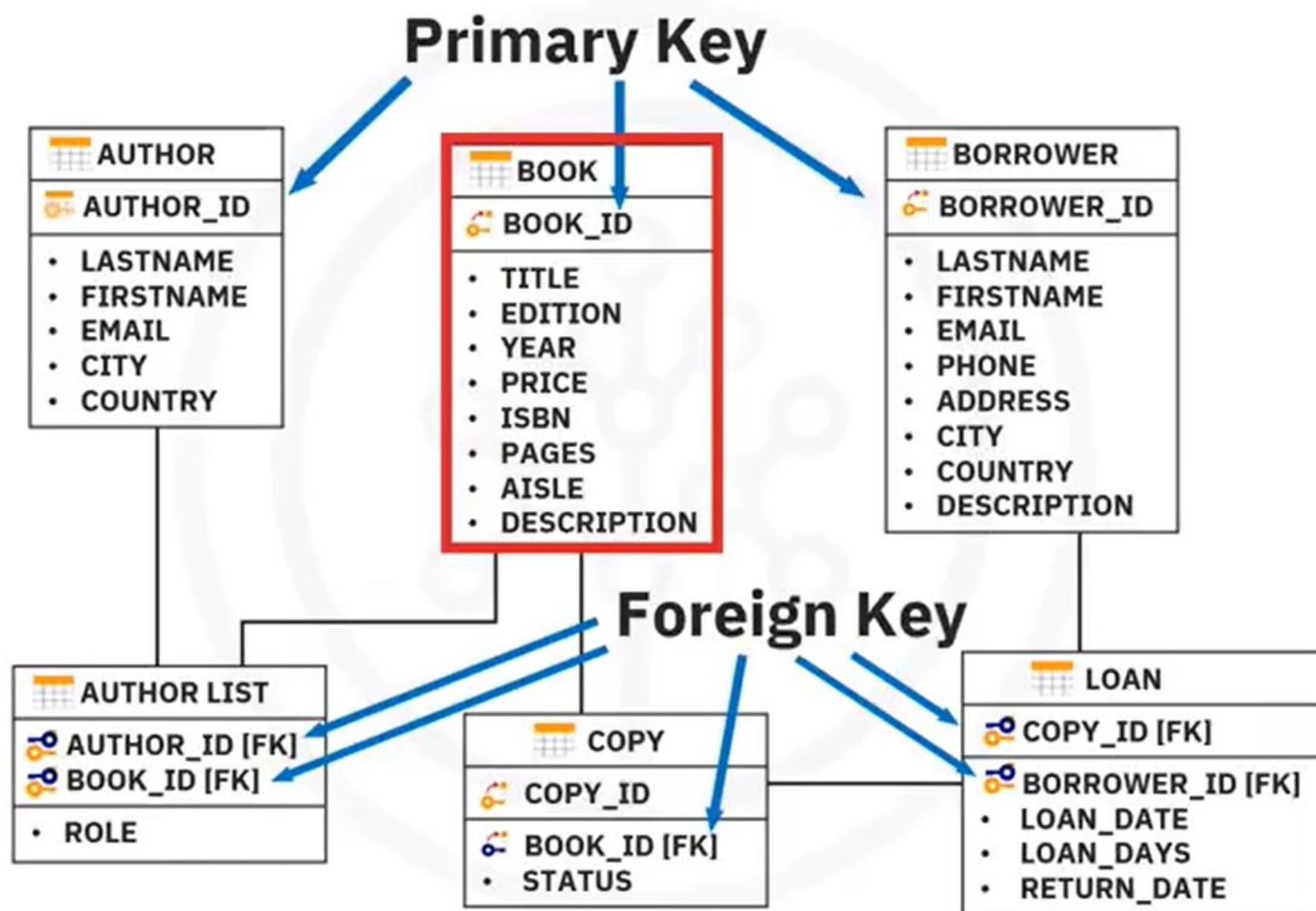
book_id	title	total_pages	rating	isbn	pub_date	pub_id
101	Lean Software Development	240	4.17	978032	2003-05-18	5
201	Facing the Intelligence Explosion	91	3.87		2013-02-01	109
301	Scala in Action	419	3.74	978194	2013-04-10	111
401	Patterns of Software	256	3.84	978020	1996-08-15	150

Creating primary keys

```
CREATE TABLE book(  
    book_id INT NOT NULL,  
    . . .  
    pub_id INT NULL,  
    PRIMARY KEY(book_id));
```

```
ALTER TABLE book  
    ADD PRIMARY KEY(book_id, ISBN);
```

What is a foreign key?

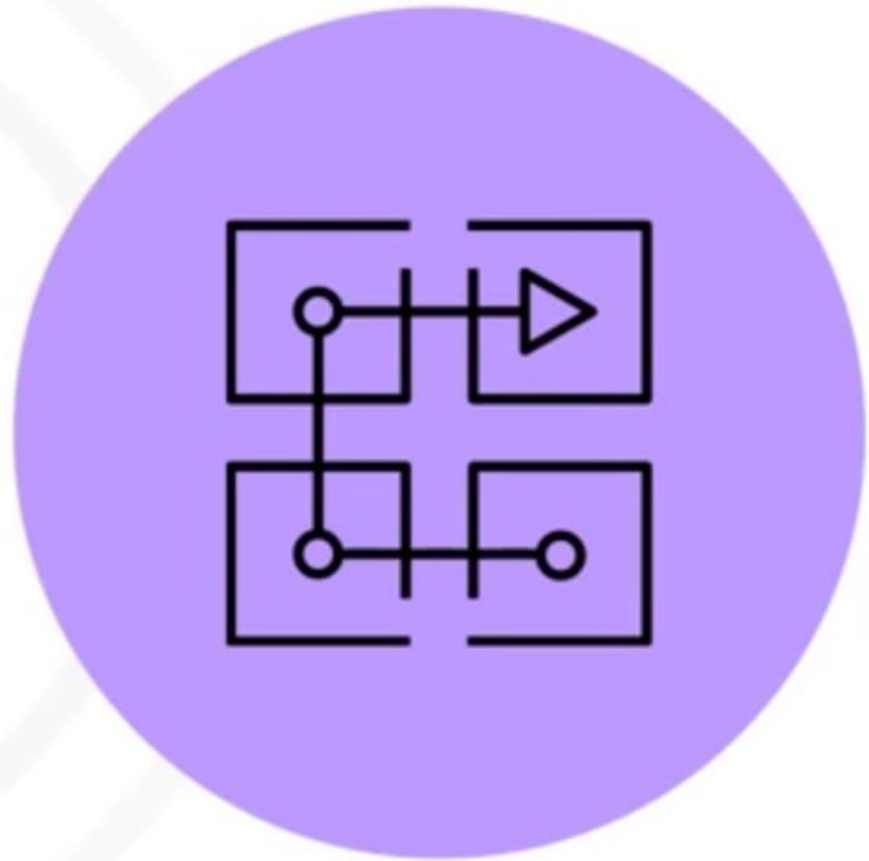


Creating foreign keys

```
CREATE TABLE copy(  
  copy_id INT NOT NULL,  
  book_id INT NULL,  
  . . .  
  CONSTRAINT fk_copy_book FOREIGN KEY(book_id)  
    REFERENCES book(book_id)  
);
```

What is normalization?

- Data duplication leads to inconsistencies
- Normalization reduces data duplication
- Increases speed of transactions
- Improves the integrity of data
- Normalizes each table
- Most used:
 - First normal form
 - Second normal form
 - Third normal form



First normal form

Each row must be unique

Each cell must contain only a single value

Also called 1NF



1NF

First normal form

Book_id	Title	Format	Author_name
101	Lean Software Development	Paperback	Mary Poppendieck
201	Facing the Intelligence Explosion	Paperback	David Robson
301	Scala in Action	Hardback	Yehuda Katz
401	Patterns of Software	Hardback, Paperback	Mary Poppendieck
501	Anatomy of LISP	Paperback	Eric Redmond

First normal form

Book_id	Title	Format	Author_name
101	Lean Software Development	Paperback	Mary Poppendieck
201	Facing the Intelligence Explosion	Paperback	David Robson
301	Scala in Action	Hardback	Yehuda Katz
401	Patterns of Software	Paperback	Mary Poppendieck
401	Patterns of Software	Hardback	Mary Poppendieck
501	Anatomy of LISP	Paperback	Eric Redmond

Second normal form

Database must be in the first normal form

Create separate tables for sets of values

Also called 2NF



2NF

First normal form

Book_id	Title	Format	Author_name
101	Lean Software Development	Paperback	Mary Poppendieck
201	Facing the Intelligence Explosion	Paperback	David Robson
301	Scala in Action	Hardback	Yehuda Katz
401	Patterns of Software	Paperback	Mary Poppendieck
401	Patterns of Software	Hardback	Mary Poppendieck
501	Anatomy of LISP	Paperback	Eric Redmond

Second normal form



Book_id (Primary Key)	Title	Author_name
101	Lean Software Development	Mary Poppendieck
201	Facing the Intelligence Explosion	David Robson
301	Scala in Action	Yehuda Katz
401	Patterns of Software	Mary Poppendieck
501	Anatomy of LISP	Eric Redmond

Book_id (Foreign Key)	Format
101	Paperback
201	Paperback
301	Hardback
401	Hardback
401	Paperback
501	Paperback

Third normal form

Database must be in the first and second normal form

Eliminate columns that do not depend on the key

Also called 3NF



3NF

Third normal form

Book_id (Primary Key)	Title	Author_name	Publisher	Ships from
101	Lean Software Development	Mary Poppendieck	Tech Books	UK
201	Facing the Intelligence Explosion	David Robson	Amazing Books	US
301	Scala in Action	Yehuda Katz	Better Tech Books	India
401	Patterns of Software	Mary Poppendieck	Publisher 101	US
501	Anatomy of LISP	Eric Redmond	Best Tech Books	Canada

Third normal form

Book_id (Primary Key)	Title	Author_name	Pub_id
101	Lean Software Development	Mary Poppendieck	1
201	Facing the Intelligence Explosion	David Robson	2
301	Scala in Action	Yehuda Katz	3
401	Patterns of Software	Mary Poppendieck	4
501	Anatomy of LISP	Eric Redmond	5

Pub_id (Primary Key)	Publisher	Ships from
1	Tech Books	UK
2	Amazing Books	US
3	Better Tech Books	India
4	Publisher 101	US
5	Best Tech Books	Canada

Normalization in OLTP and OLAP

- Online transactional processing (OLTP)
 - Involves frequent reading and writing of data
 - Normalizes data to 3NF
- Online analytical processing (OLAP)
 - Primarily deals with read-only data